

Qatar-10b

R-0084

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_231026 · created 2026-07-15T20:57:35+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460243.6733 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.012
Transit depth (Rp/Rs)^2	3.02 +/- 0.42 [%]
Orbital Inclination (inc)	89.34 +/- 2.42
Airmass coefficient 1 (a1)	1.017 +/- 0.015
Airmass coefficient 2 (a2)	-0.008 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	#3 - [460, 84]
Optimal Method	PSF photometry
Transit Duration (day)	0.0483 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.174 ± 0.012 vs 0.1265 (deviation 2.93σ)
Duration fit vs published	0.0483 d vs 0.1154 d (deviation 10.58σ)
Mid-time O-C	-21.8 min
Depth SNR	10.7
Residual scatter	1.47 %

Light curve

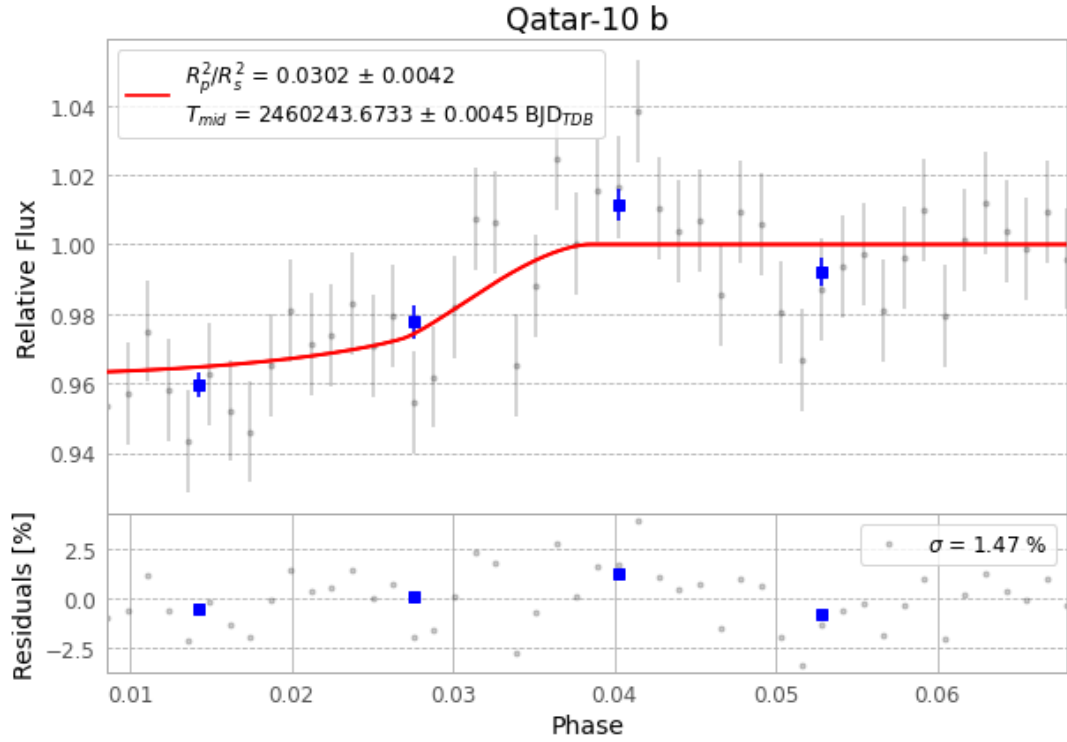


Figure 1 — FinalLightCurve_Qatar-10 b_2023-10-25.png (SHA-256 89edd78dcc2f19ce...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 908150b41a93bf61...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

48 FITS frames (32 MB), Qatar-10231026042715.FITS → Qatar-10231026064816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-231026.log (2db9109cc35e...), FinalLightCurve_Qatar-10 b_2023-10-25.png (89edd78dcc2f...), FinalLightCurve_Qatar-10 b_2023-10-25.csv (bc8f96d26877...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 20:57Z.